

Raunak Kumar

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<https://ca.linkedin.com/pub/raunak-kumar/94/12b/879>

<https://raunakkmr.github.io>

SUMMARY I am passionate about algorithms, machine learning, systems and competitive programming. I take keen interest in machine learning research and teaching.

EDUCATION *B.Sc. Computer Science,* September 2013 - May 2018
University of British Columbia, Vancouver, BC

- Grade in Computer Science and Mathematics courses: 90.4%
- Related Coursework:
 - Machine Learning (graduate course), Advanced Algorithm Design and Analysis, Randomized Big Data Algorithms (graduate course), Theory of Computation
 - Probability, Real Analysis, Graph Theory, Linear Algebra, Linear Programming
 - Internet Computing, Advanced Operating Systems, Databases

AWARDS J. Fred Muir Memorial Scholarship in Science ('13-'14)
Trek Excellence Scholarship ('13-'14) - Top 5% of undergraduates
Science Scholar / Dean's Honors List ('13-'14, '14-'15, '15-'16, '16-'17)
UBC Computer Science Teaching Assistant Award ('16-'17)

TECHNOLOGY SKILLS *Programming Languages:* C++, Python, C, Matlab, Java.
Software: TensorFlow, PyTorch, Numpy, Scikit-Learn, Vim, Git.
Operating Systems: Linux Ubuntu, Windows.

EXPERIENCE *NSERC Research Assistantship* May 2017 - Present
Computer Science Department, UBC

- Working on theoretical and applied machine learning problems under the supervision of Prof. Mark Schmidt.
- We give the first convergence rate for non-convex functions in the surrogate optimization framework. We view expectation-maximization (EM) in this framework and, consequently, also provide a convergence rate for EM. We are also working on using other ideas from optimization to improve EM.
- Along with a graduate student, I am also working on improving deep learning architectures for image segmentation.

Software Engineering Intern May 2016 - August 2016
Google, Mountain View, CA

- Worked on a distributed datastore trigger utility in C++ for the App and Product Infrastructure team. Triggers are a mechanism by which actions can be scheduled for the future. The goal of the utility is to provide an easy and extensible way to maintain and debug triggers.

TEACHING

Instructor / Coordinator

January 2017 - April 2017

Computer Science Department, UBC

- Co-taught CPSC 490: Problem Solving in Computer Science, focusing on practical applications of advanced algorithms and data structures.
- Responsibilities included lecturing a class of 15-30 students 3 times a week, writing lecture notes and writing assignments. Website: www.ugrad.cs.ubc.ca/~cs490/2016W2/.

Teaching Assistant

September 2014 - Present

Computer Science Department, UBC

- Worked as a TA for 8 terms for the following courses: Advanced Algorithm Design and Analysis, Machine Learning, Advanced Operating Systems, Intermediate Algorithm Design and Analysis, Introduction to Computer Systems, Models of Computation. I was an undergrad when I worked as a TA for the Advanced Algorithms and Machine Learning courses. Usually, only grad students are assigned to TA these courses.
- Responsibilities included lecturing sections ranging from 15 - 100 students, holding office hours and grading. I have taught an entire lecture for two algorithms courses as well.

PROJECTS

Programming Competitions

- UBC ACM Team: 2015 - Present.
- UBC ACM Team Coach (Div. 2) - 2017 - Present.
- ACM-ICPC PacNW Regionals: 15th (2016, Div. 1), 6th (2015, Div. 2).
- Microsoft Coding Competition MSFT3C: 2nd (2017), 6th (2016).
- SFU Winter Programming Contest: 8th (2017).
- NAIPC (Open Division USA/Canada): 27th (2016).
- Calgary Collegiate Programming Competition: 3rd (2015, Div. 2).

Approximating Steiner Trees in the Semi-Streaming Model

- Presented a $2(1+\epsilon)(2^{\frac{\log n}{\log \log n}}+1)$ -approximation for the weighed metric Steiner tree problem using $O(n \text{ polylog} n)$ space, as part of a course project for a graduate class on randomized big data algorithms. Paper can be found on my website.

Machine Learning Reading Group

- I present at the Machine Learning Reading Group and the Machine Learning Theory Reading Group. I have presented on topics on learning theory, reinforcement learning and online learning. Slides can be found on my website.

Gas Detection

- Implemented random forest and neural network classification to detect dangerous gases based on sensor measurements for a company in Victoria, BC. Also implemented visualization of the training data using PCA and ISOMAP.

Kernel

- Implemented basic functionality of a kernel using C and x86 assembly on the Bochs emulator. Features included memory management, process management, system calls, interrupt handling, signals and hardware interrupts.

Web Proxy Server

- Created a multithreaded server application using C and Unix sockets to access HTTP content from websites, block blacklisted websites and cache results.

Dictionary Client

- Created a client side application using Java and TCP to connect with specified dictionary servers and retrieve the requested definitions.

REFERENCES

Mark Schmidt

- Research supervisor and professor for machine learning courses.
- email: schmidtm@cs.ubc.ca, phone: 604-822-6421

William Evans

- Professor for Advanced Algorithms course, worked as a TA under him for the same, and faculty sponsor for the problem solving course I taught.
- email: schmidtm@cs.ubc.ca, phone: 604-822-0827

Nicholas Harvey

- Professor for Randomized Big Data Algorithms and Theory of Computation courses.
- email: nickhar@cs.ubc.ca, phone: 604-822-5491

OTHER

Canadian citizen.